Introduction into Possibilistic Logic

Possibilistic Logic - An Overview by Didier Dubois and Henri Prade

Alexander Köhn

Artificial Intelligence Group,
University of Hagen, Germany

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Overview

What is Possibilistic Logic?

- **Definition**: Possibilistic logic is a weighted logic that handles uncertainty in a qualitative way by adding certainty to classical logic formulas
- **Applications**: Used in expert systems, decision-making, and AI for reasoning under uncertainty.

Overview

What are Possibility Distributions?

- A **possibility distribution** is a mapping:

$$\pi : U \to S,$$

  where:
  - U: Set of states
  - S: Totally ordered scale ($S = [0, 1]$ or finite chains).
  - $\pi(u)$: Degree of possibility (plausibility) of state u.
- $\pi(u) = 0$: State u is **impossible**.
- $\pi(u) = 1$: State u is **fully possible**.

Normalization Condition

At least one state $u \in U$ must have $\pi(u) = 1$ (an actual state exists).

Specificity of a Possibility Distribution

- A distribution π is **at least as specific** as π' if:

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

- Examples:
  - **Complete knowledge:** $\pi(u_0) = 1$, $\pi(u) = 0$ for all $u \neq u_0$.
  - **Complete ignorance:** $\pi(u) = 1$ for all $u \in U$.
- Principle of minimal specificity: It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

Possibility and Necessity Functions

- Possibility measure:
$$\Pi(A) = \sup_{u \in A} \pi(u).$$
Evaluates the extent to which A is consistent with π.
- Necessity measure:
$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$
Evaluates the extent to which A is certainly implied by π.

Duality Relationship

$$N(A) = 1 - \Pi(A^c).$$

Where A^c is the complement of A.

Key Properties of Measures

- **Possibility: Maxitivity Property**

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)).$$

- **Necessity: Minimization Property**

$$N(A \cap B) = \min(N(A), N(B)).$$

Certainty Qualification in Possibilistic Logic

- Certainty-qualified information: "A is certain to degree α" is modeled by the constraint:

$$N(A) \geq \alpha$$

- Least Specific Possibility Distribution:

$$\pi_{(A,\alpha)}(u) = \begin{cases} 1 & \text{if } u \in A, \\ 1 - \alpha & \text{otherwise.} \end{cases}$$

  - If $\alpha = 1$: $\pi(A, \alpha)$ becomes the characteristic function of A.
  - If $\alpha = 0$: $\pi(A, \alpha)$ represents total ignorance.

Example

- A doctor is trying to diagnose a patient based on symptoms and diseases:
  - $U = \{\text{flu}, \text{cold}, \text{allergy}\}$.
  - Possibility distribution π based on symptom severity:

$$\pi(\text{flu}) = 0.9, \quad \pi(\text{cold}) = 0.6, \quad \pi(\text{allergy}) = 0.3.$$

- The doctor evaluates the possibility and necessity measures for a hypothesis A:

$$A = \{\text{flu}, \text{cold}\}$$

- Calculation:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(\text{flu}), \pi(\text{cold})) = 0.9.$$

$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(\text{allergy}) = 1 - 0.3 = 0.7.$$

- Interpretation:
  - Possibility ($\Pi(A)$): Flu or cold is highly possible (0.9).
  - Necessity ($N(A)$): It is moderately certain (0.7) that the patient has either flu or cold.

Overview

Possibilistic Logic

- A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

  where:
  - a: A classical logic formula (e.g., $p \to q$).
  - $\alpha \in [0, 1]$: Certainty level (minimum necessity degree of a).
- Interpretation:
  - (a, α): "Formula a is at least certain to degree α."
  - The higher α, the stronger the belief in a.

- Certainty weakening:

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta)$$

Example

(It will rain tomorrow, 0.9) $\vdash$ (It will rain tomorrow, 0.7) - If the certainty level decreases, the conclusion still holds.

- Modus ponens:

$$(\neg a \vee b, \alpha), \quad (a, \alpha) \vdash (b, \alpha)$$

Example

$(\neg\text{fever} \vee \text{flu}, 0.8), \quad (\text{fever}, 0.8) \vdash (\text{flu}, 0.8)$ - If fever is observed, flu is inferred with certainty 0.8.

- Resolution:

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha)$$

Example

$(\neg\text{fever} \vee \text{flu}, 0.8), \quad (\text{fever} \vee \text{headache}, 0.8) \vdash (\text{flu} \vee \text{headache}, 0.8)$

- Weakest link resolution:

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta))$$

Example

$(\neg\text{fever} \vee \text{flu}, 0.9), \quad (\text{fever} \vee \text{headache}, 0.7) \vdash (\text{flu} \vee \text{headache}, 0.7)$ - The weakest certainty (0.7) determines the conclusion.

- Formula weakening:

$$\text{If } a \vdash b \text{ in classical logic, then } (a, \alpha) \vdash (b, \alpha).$$

Example

$(\text{fever}, 0.9) \vdash (\text{flu}, 0.9)$ - Logical implication is preserved with the same certainty level.

- Possibilistic logic can handle conflicting information by:
  - Identifying an **inconsistency level** ($inc - l$).
  - Filtering out rules with certainty levels below $inc - l$.
- **Example**:

$$\{(a, 0.8), (\neg a, 0.7)\}$$

  - Inconsistency level: $inc - l = 0.7$.
  - Filtered result: $(a, 0.8)$ dominates.

Logical Base and Weighted Clauses

- A possibilistic logic base is a collection of weighted formulas:

$$\{(a_1, \alpha_1), (a_2, \alpha_2), \dots\}.$$

- Clauses can be rewritten in terms of their weights:
  - Example:

$$(a \wedge b, \alpha) \quad \Leftrightarrow \quad \{(a, \alpha), (b, \alpha)\}.$$

- Logical tautologies (t) hold with any certainty level:

$$(t, \alpha), \quad \forall \alpha \in [0, 1].$$

Overview

Example: Medical Diagnosis

- Rule base:

$$(\neg \mathsf{fever} \vee \mathsf{flu}, 0.9),$$
$$(\neg \mathsf{cough} \vee \mathsf{cold}, 0.8),$$
$$(\mathsf{fever} \vee \mathsf{headache}, 0.7).$$

- Facts:

$$(\mathsf{fever}, 1), \quad (\mathsf{cough}, 1).$$

- Inference:
  - From Modus Ponens:

$$(\mathsf{flu}, 0.9), \quad (\mathsf{cold}, 0.8).$$

  - From Resolution:

$$(\mathsf{flu} \vee \mathsf{headache}, 0.7).$$

Overview

Conclusion

- Possibilistic logic provides a robust framework for reasoning under uncertainty.
- Key features:
  - Certainty levels for qualitative inference.
  - Compatibility with classical logic.
  - Handling of conflicting information.
- Applications:
  - Medical diagnosis, decision-making, and knowledge representation.

Overview

Basic Syntax

- A possibilistic formula is a pair:

$$(a, \alpha), \quad \text{where } \alpha \in [0, 1].$$

- Interpretation:
  - a: A propositional formula.
  - α: Certainty level (degree of belief in a).
- **Examples**:
  - $(b \rightarrow f, 0.9)$: Birds fly with certainty 0.9.
  - $(p \rightarrow \neg f, 1)$: Penguins do not fly, absolutely certain.

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Key Idea

Certainty levels enrich classical logic, enabling reasoning about typical cases and exceptions.

Key Inference Rules

- **Certainty Weakening**:

$$\underline{\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta)}.$$

- **Modus Ponens**:

$$\underline{(\neg a \vee b, \alpha), \quad (a, \alpha) \vdash (b, \alpha)}.$$

- **Resolution**:

$$\underline{(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha)}.$$

- **Weakest Link Resolution**:

$$\underline{(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta))}.$$

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Penguin Example

Consider:

$$(\neg b \vee f, 0.9), \quad (\neg p \vee \neg f, 1), \quad (\neg p \vee b, 1).$$

Overview

Default Rules as Constraints

- Default rule:
$$\underline{\Pi(a \wedge b) > \Pi(a \wedge \neg b)}.$$
- $a \to b$: "If a, then b is more plausible than $\neg b$."
- In terms of necessity:
$$\underline{N(b|a) = 1 - \Pi(\neg b|a) > 0}.$$
- Comparative relation:
$$\underline{a \wedge b >_{\Pi} a \wedge \neg b}.$$

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Penguin Example Constraints

$$\begin{aligned} b \wedge f >_{\Pi} b \wedge \neg f & \quad \text{(birds fly)}. \\ p \wedge \neg f >_{\Pi} p \wedge f & \quad \text{(penguins don't fly)}. \\ p \wedge b >_{\Pi} p \wedge \neg b & \quad \text{(penguins are birds)}. \end{aligned}$$

Overview

Computing Necessity for Penguin Example

- **Possibility Distribution**:

$$\pi(\omega) \text{ values assigned to worlds } \omega.$$

- Worlds:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

Computing Necessity for Penguin Example

- **Possibility Distribution**:

$$\pi(\omega) \text{ values assigned to worlds } \omega.$$

- Worlds:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

- **Constraints**:

$$C_1 : \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)).$$
$$C_2 : \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)).$$
$$C_3 : \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).$$

Inference with New Information

- Factual knowledge:

$$(b, 1) \quad \text{(Tweety is a bird)}.$$

- Possibilistic inference:

$$K \cup \{(b, 1)\} \vdash (f, 1/3).$$

  - Default rule: Tweety flies with certainty $1/3$.

- Adding $p = 1$ (Tweety is a penguin):

$$K \cup F \vdash (\neg f, 2/3).$$

  - New inference: Tweety does not fly.

Overview

Conclusion

- Possibilistic logic effectively models:
  - Default reasoning with exceptions.
  - Reasoning under uncertainty.
- **Penguin Example**:
  - Illustrated how rules like "birds fly" and "penguins don't fly" can be reconciled.
  - Demonstrated inference and conflict resolution.
- **Applications**:
  - Knowledge bases.
  - Default rule modeling.
  - Nonmonotonic reasoning.

Overview

Applications of Possibilistic Logic

- **Medical Diagnosis**:
  - Rules with certainty for differential diagnoses.
- **Decision-Making**:
  - Selecting actions under incomplete knowledge.
- **Knowledge Bases**:
  - Handling updates with belief revision.

Example Wrap-Up

Throughout the presentation, we used:

- Syntax: Representation of rules ((a, α)).
- Measures: Calculating $\Pi(A)$ and $N(A)$.
- Inference: Combining uncertain rules.
- Inconsistencies: Managing conflicts in rules.

- **Default Reasoning**: Handling incomplete knowledge in expert systems.
- **Medical Diagnosis**: Prioritizing plausible diagnoses based on symptoms.
- **Decision-Making**: Evaluating options under uncertainty.
- **Fuzzy Control Systems**: Representing and processing vague information.

Overview

Overview

Possibility distribution:

U is a set of affairs.

A possibility distribution is a mapping π from U to a totally ordered scale S, with top denoted by 1 and bottom by 0. In the finite case $S = \{1 = \lambda_1 > ...\lambda_n > \lambda_{n+1} = 0\}$.

The function π represents the state of knowledge of an agent (about the actual state of affairs), also called an epistemic state.

$\pi(u) = 0$ means that state u is rejected as impossible;

$\pi(u) = 1$ means that state u is totally possible ($=$ plausible).

Specificity: A possibility distribution π is said to be at least as specific as another π' if and only if for each state of affairs u: $\pi(u) \leq \pi'(u)$. Then, π is at least as restrictive and informative as π' , since it rules out at least as many states with at least as much strength.

Complete knowledge: for some u_0, $\pi(u_0) = 1$ and $\pi(u) = 0$, $\forall u \neq u_0$ (only u_0, is possible).

Complete ignorance: $\pi(u) = 1, \forall u \in U$ (all states are possible).

Principle of minimal specificity: It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

Possibility and necessity functions: $\Pi(A) = \sup_{u \in A} \pi(u)$; $\quad N(A) = \inf_{u \notin A}(1 - \pi(u))$

Overview

Blocks

Block

This is a regular block.

- This is an item in a block.

Block

This is an alert block.

- This is an item in an alert block.

Block

This is an example block.

- This is an item in an example block.

Highlighting Text

We can highlight text. There are also a darker option, and a yellow option.

The same options are also available in math mode: $a + b = c$

This is an appendix.

- Page numbers start from the beginning.